

JAEHOON HAHM

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RESEARCH INTEREST

- AI + Quantum Physics, Machine Learning for Many-body Physics, Neural Quantum States
- Riemannian Geometry + Flow/Diffusion Models, Geometric Deep Learning

EDUCATION

University of Illinois, Urbana-Champaign (UIUC)

Aug 2024 - Present

Ph.D. student in Physics

- Advisor: Prof. Bryan Clark
- Machine Learning for Many-body Physics, Neural Quantum States, Generative Modeling of Quantum State

Graduate School of Data Science, Seoul National University

Mar 2022 - Feb 2024

M.S. in Data Science

- Thesis: Isometric Representation Learning for Disentangled Latent Space of Diffusion Models
- Advisor: Prof. Joonseok Lee

Seoul National University

Mar 2016 - Feb 2022

B.S. in Physics, minor in Mathematics

- Thesis: Improvement in Variational Quantum Algorithms by Measurement Simplification
- Advisor: Prof. Young June Park
- Mandatory Military Service at the Republic of Korea Army: July 2019 - Feb 2021

Seoul Science High School

Mar 2013 - Feb 2016

Graduated with Distinction

- Science and Mathematics for Gifted Students

PUBLICATION

1. **Jaehoon Hahm**, Bryan Clark, "Shadow Flow Matching for Learning Quantum States", Under Review at NeurIPS, 2025.
2. Kwanseok Kim, **Jaehoon Hahm**, Jinhwan Sul, and Joonseok Lee, "SummDiff: Generative Modeling of Video Summarization with Diffusion", International Conference on Computer Vision (ICCV), 2025.
3. **Jaehoon Hahm**, Tak Hur, Joonseok Lee, Daniel K. Park, "Generative Modeling of Quantum Distribution with Functional Flow Matching", Quantum Techniques in Machine Learning (QTML), 2024.
4. **Jaehoon Hahm**, Junho Lee, Sunghyun Kim, and Joonseok Lee, "Isometric Representation Learning for Disentangled Latent Space of Diffusion Models", International Conference on Machine Learning (ICML), 2024.
5. **Jaehoon Hahm**, Hayeon Kim, and Young June Park, "Improvement in Variational Quantum Algorithms by Measurement Simplification", *preprint* arXiv:2312.06176.

EXPERIENCE

Undergraduate Research Intern

June 2018 - April 2019, Feb 2021 - Dec 2021

Advisor: Prof. Young June Park

Department of Electrical & Computer Engineering, SNU

- **Improvement in Variational Quantum Algorithms by Measurement Simplification:**
Research project of improving Variational Quantum Algorithms by utilizing only partial information of the state vector. Reviewed simple rules in quantum circuits, and propose a simplification method, Measurement

Simplification, that simplifies the expression for the measurement of quantum circuit. The research was part of and supported by Samsung *Beyond Limit* project.

- **Gate Level Implementation of Quantum KNN:**

Leader of Quantum Algorithm and Quantum Machine Learning research team. Investigated about Quantum KNN algorithm and implemented it via IBM Qiskit that can be executed at IBMQ real qubit hardware. The research was part of and supported by Samsung *Beyond Limit* project.

Undergraduate Research Intern

Feb 2021 - Dec 2021

Advisor: Prof. Taehyun Kim

Department of Computer Science & Engineering, SNU

- **Randomized Benchmarking of Trapped Ion Qubit:**

Research project of measuring the performance of Trapped Ion Qubit in the lab via IBM Qiskit and Randomized Benchmarking.

Undergraduate Research Intern

July 2021 - Dec 2021

Advisor: Prof. Hyunyoung Choi

Department of Physics, SNU

- **Diamond NV center Control Experiment:**

Diamond NV center pulse control experiment, including sequences of Rabi oscillation, spin echo, dynamical decoupling.

Mandatory Military Service at the Republic of Korea Army

July 2019 - Feb 2021

Chemical, Biological, Radiological and Nuclear weapons Reconnaissance, 5th Infantry Division

- 19 months of mandatory military service, Republic of Korea Army.

LANGUAGES

Programming Languages

Python, C, C++, JAVA, MATLAB, Mathematica

Frameworks

PyTorch, JAX, TensorFlow, Numpy, Pandas

Quantum Languages

Qiskit, Cirq, PennyLane, TensorFlow Quantum

Languages

English (Fluent), Korean (Native)